25/5/15

# Quadratic Equation

## Standard form of Quadratic Equation:

$f(x) = 0$

where $f(x)$ is quadratic equation

$ax^2 + bx + c = 0$

LHS ↓ RHS

Degree is 2

For example:

(i) $3x^2 = 0$

(ii) $3x^2 + 4x + 7 = 3x^2$

$4x + 7 = 0$

(iii) $(x+1)^3 = x^2$

$x^3 + 1 + 1 + 3x^2 + 3x = x^2$

$2x + 1$

## Finding Solutions of Quadratic Equation

- Factorization Method
  - By regrouping
  - By splitting the terms
  - By using Identity
- Quadratic Formula

### ① By regrouping:

$\Rightarrow x^2 + 5x + 6 = x^2 + 2x + 3x + 6$

$\Rightarrow 3x(x+2) + 3(x+2)$

$\Rightarrow (x+3)(x+2)$

### ② By splitting the middle term:

$\Rightarrow 2x^2 + 5x + 3$

$\Rightarrow 2x^2 + 2x + 3x + 3$

$\Rightarrow 2x(x+1) + 3(x+1)$

$\Rightarrow (2x+3)(x+1)$

### ③ By using Identities:

$\Rightarrow x^2 + 4x + 4 = 0$

$\Rightarrow (x)^2 + 2(2)(x) + (2)^2 = (x+2)^2$

$\Rightarrow (x+2)^2 = 0$

$\Rightarrow (x+2)(x+2) = 0$

$\Rightarrow 9x^2 - 16$

$\Rightarrow (3x)^2 - (4)^2 = (3x+4)(3x-4) = 0$

$\Rightarrow x^2 - \frac{1}{100} = 0$

$\Rightarrow (x)^2 - \left(\frac{1}{10}\right)^2 = \left(x - \frac{1}{10}\right)\left(x + \frac{1}{10}\right) = 0$

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## Exercise - 4.1

i-ii-Sol. $\Rightarrow x^2 - 2x = (2)(3-x)$ — ①

$\Rightarrow x^2 - 2x = -6 + 2x$

$\Rightarrow x^2 - 2x + 6 - 2x = 0$

$\Rightarrow x^2 - 4x + 6 = 0$

which is of the form $ax^2+bx+c$ ; $a \neq 0$

Hence, ① is a quadratic equation.

iv-Sol. $\Rightarrow (x-2)(2x+1) = x(x+5)$ — ①

$\Rightarrow 2x^2 + x - 4x - 2 = x^2 + 5x$

$\Rightarrow 2x^2 - 3x - 2 - x^2 - 5x = 0$

$\Rightarrow x^2 - 10x - 2 = 0$,

which is of the form $ax^2+bx+c$ ; $a \neq 0$

Hence, ① is a quadratic equation.

vi-Sol. $\Rightarrow x^2 + 3x + 1 = (x-2)^2$ — ①

$\Rightarrow x^2 + 3x + 1 = x^2 + 4 - 4x$

$\Rightarrow 7x - 3 = 0$

which has no term containing $x^2$

Hence, ① is not a quadratic equation.

viii-Sol. $\Rightarrow x^3 - 4x^2 - x + 1 = (x-2)^3$ — ①

$\Rightarrow x^3 - 4x^2 - x + 1 = x^3 - 8 - 6x^2 + 12x$

$\Rightarrow 2x^2 - 13x + 9 = 0$

which is of the form $ax^2+bx+c = 0$, $a \neq 0$

Hence, ① is a quadratic equation.

2-ii-Sol. Let '$x$' and '$x+1$' be two consecutive positive integers.

∴ Product of two integers $= x(x+1) = x^2 + x$

By the question, $x^2 + x = 306$

$\Rightarrow x^2 + x - 306 = 0$

$\Rightarrow x^2 + 18x - 17x - 306 = 0$

$\Rightarrow x(x+18) - 17(x+18) = 0$

$\Rightarrow (x-17)(x+18) = 0$

$\Rightarrow x+18 = 0$ or $x - 17 = 0$

$\Rightarrow x = -18$ or $x = 17$

Thus, $x = 17$ [∵ $x > 0$]

Hence, the two consecutive integers are 17 and 17+1 i.e, 17 and 18. And the quadratic equation is $x^2 + x - 306 = 0$

## Exercise 4.2

1-i-Sol. $\Rightarrow x^2 - 3x - 10 = 0$ — ①

$\Rightarrow x^2 - 5x + 2x - 10 = 0$

$\Rightarrow x(x-5) + 2(x-5) = 0$

$\Rightarrow (x+2)(x-5) = 0$

$\Rightarrow x - 5 = 0$ or $x + 2 = 0$

$\Rightarrow x = 5$ $x = -2$

Hence, 5 and $-2$ are the roots for ①.

iii-Sol. $\Rightarrow \sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ — ①

$\Rightarrow \sqrt{2}x^2 + 2x + 5x + 5\sqrt{2} = 0$

$\Rightarrow \sqrt{2}x(x+\sqrt{2}) + 5(x+\sqrt{2}) = 0$

$\Rightarrow (x+\sqrt{2})(\sqrt{2}x+5) = 0$

$\Rightarrow x + \sqrt{2} = 0$ or $\sqrt{2}x + 5 = 0$

$\Rightarrow x = -\sqrt{2}$ $x = -\frac{5}{\sqrt{2}}$

Hence, the roots are $-\sqrt{2}$ and $-\frac{5}{\sqrt{2}}$ for ①.

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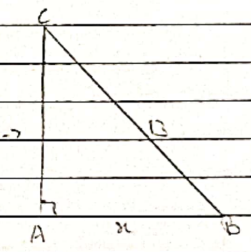
2-i-Sol. Let 'x' be the number of marbles John had.
∴ no. of marbles Jivanti had = 45 − x.
When each lost 5 marbles, then the number of marbles left with John = x − 5 and the number of marbles left with Jivanti = (45 − x) − 5 = 40 − x.
∴ Product = (x − 5)(40 − x)
= 40x − x² − 200 + 5x = 45x − x² − 200
By the question, −x² + 45x − 200 = 124
x² − 45x + 324 = 0
x² − 36x − 9x + 324 = 0
x(x − 36) − 9(x − 36) = 0
(x − 36)(x − 9) = 0
⇒ x − 36 = 0 or x − 9 = 0
⇒ x = 36 or x = 9
Hence, the no. of marbles they had to start with are 36, 9.

3-Sol. Let one number be x
Then the other number = 27 − x
∴ Product = x(27 − x) = 27x − x²
By the question, 27x − x² = 182
−x² + 27x − 182 = 0
−x² + 13x + 14x − 182 = 0
x² − 13x − 14x + 182 = 0
x(x − 13) − 14(x − 13) = 0
(x − 14)(x − 13) = 0
⇒ x − 14 = 0 or x − 13 = 0
⇒ x = 14 or x = 13
Hence, the two numbers are 13 and 14.

4-Sol. Let first integer be x
∴ 2nd Integer = x + 1
According to question
x² + (x + 1)² = 365
x² + x² + 2x = 365
2x² + 2x = 365
2x² + 2x − 364 = 0
x² + x − 182 = 0
x² + 14x − 13x − 182 = 0
x(x + 14) − 13(x + 14) = 0
(x − 13)(x + 14) = 0
x = −14; x = 13

| x = −14 | x = 13 |
|---|---|
| x + 1 = −14 + 1 | x + 1 = 13 + 1 = 14 |
| = −13 | = 13, 14 |
| Not applicable | |

5-Sol. In △ABC, ∠A = 90
using Pythagoreas theorem




BC² = AC² + AB²
(13)² = (x − 7)² + x²
169 = x² + 49 − 14x + x²
120 = 2x² − 14x
0 = 2x² − 14x − 120
0 = 2(x² − 7x − 60)
0 = x² − 12x + 5x − 60
0 = x(x − 12) + 5(x − 12)
0 = (x − 12)(x + 5)
AB = 12 AC = 12 − 7 = 5

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6-Sol Let 'x' be the number of poetry articles produced by cottage industary in a day.

∴ Cost of production of each article = ₹(2x+3)

∴ Total cost of production in a day = ₹x(2x+3)

= ₹$2x^2+3x$

By the question, $2x^2+3x = 90$

$2x^2+3x-90=0$

$2x^2-12x+15x-90=0$

$2x(x-6)+15(x-6)=0$

$(2x+15)(x-6)=0$

$\Rightarrow 2x+15=0$ or $x-6=0$

$\Rightarrow x=-\frac{15}{2}$ or $x=6$

Thus, $x=6$

Hence, The number of articles produced in a day = 6 and cost of production of each article

= ₹(2×6+3) = ₹15

## Nature of Roots

$ax^2+bx+c=0$

- $D>0$ → Roots are real & Distinct
- $D=0$ → Roots are equal.
- $D<0$ → Roots are not real.

$D=b^2-4ac$

## Exercise - 4.3

1-i-Sol $\Rightarrow 2x^2-3x+5=0$

$\Rightarrow ax^2+bx+c=0$

$\Rightarrow a=2,\ b=-3,\ c=5$

$D=(-3)^2-4(2)(5)$

$= 9-40 = -31<0$

Roots are not real

ii-Sol $\Rightarrow 3x^2-4\sqrt{3}x+4=0$

$\Rightarrow ax^2+bx+c=0$

$\Rightarrow a=3,\ b=-4\sqrt{3},\ c=4$

$\Rightarrow D=(-4\sqrt{3})^2-4(3)(4)$

$= 48-48 = 0$

Roots are real

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iii-Sol. $\Rightarrow 2x^2 - 6x + 3 = 0$

$\Rightarrow ax^2 + bx + c = 0$

$\Rightarrow a = 2,\ b = -6,\ c = 3$

$\Rightarrow D = (-6)^2 - 4(2)(3)$

$= 36 - 24 = 12 > 0$

Roots are real

1-i-Sol. $\Rightarrow 2x^2 + Kx + 2 = 0$

Condition Given: For equal rights

$D = 0$

$b^2 - 4ac = 0$

$K^2 - 4(2)(2) = 0$

$K^2 - 24 = 0$

$K^2 = 24$

$K = \pm\sqrt{24}$

$K = \pm 2\sqrt{6}$

ii-Sol. $\Rightarrow Kx(x-2) + 6 = 0$

$\Rightarrow Kx^2 - 2Kx + 6 = 0$

Condition Given: For equal rights

$D = 0$

$b^2 - 4ac = 0$

$(-2K)^2 - 4(1)(6) = 0$

$4K^2 - 48 = 0$

$4K(K - 6) = 0$

$4K = 0$ | $K - 6 = 0$

$K = \frac{0}{4}$ | $K = 6$

NA | $\therefore K = 6$

3-Sol. Let breadth be $x$

Length $= 2x$

Area $= 800$

$2x \times x = 800$

$x^2 = 400$

$x = 20\ m^2$

$x^2 - 400 = 0$

$D = (0)^2 - 4(1)(-400)$

$D = 0 + 1600$

$= 1600 > 0$

$\therefore$ The roots are real

4-Sol. Let the ages be $x$ and $y$

So, $x + y = 20$ ... (1)

$x + y + 4 + 4 = 48$

$\therefore x + y = 40$ ... (1)

$x + y + 8 = 48$ ... (2)

$\Rightarrow$ Putting eq. (1) in eq. (2)

$\Rightarrow 20 + 8 = 48$

$\Rightarrow 28 \neq 48$

$\therefore$ This

Let the age of first friend be $x$

$\therefore$ Age of Second friend $= 20 - x$

Four years Ago,

Age of first friend $= x - 4$

Age of Second friends $= 20 - x - 4$

$= 16 - x$

Product $= (x - 4)(16 - x)$

$= 16x - x^2 - 64 + 4x$

$= -x^2 + 20x - 64$

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By the question,

$-x^2+20x-64 = 48$

$\Rightarrow -x^2+20x-64-48=0$

$\Rightarrow -x^2+20x-112=0$

$\Rightarrow x^2-20x+112=0$ — (1)

Comparing with $ax^2+bx+c=0$, we have:

$a=1, b=-20, c=112$

$\therefore D = b^2-4ac$

$=(-20)^2-4(1)(112)$

$=400-448$

$D=-48<0$

Roots are not real

Ans. Hence, the given situation is not possible.

5-Sol. Let x and y be length and breadth of rectangular park.

$\therefore$ Perimeter $= 2(x+y)$ m

area $= xy$ m$^2$

Given,

$2(x+y)=80$ and $xy=400$

$x+y=40$ — (1)

$xy=400$ — (2)

From (1), $x=40-y$ — (3)

Putting value of x in (2)

$x(40-x)=400$

$40x-x^2=400$

$0=400+x^2-40x$

$0=x^2-40x+400$ — (4)

comparing with $ax^2+by+c=0$, we have:

$a=1, b=-40, c=400$

$D=b^2-4ac$

$D=(-40)^2-4(1)(400)$

$D=1600-1600$

$D=0=0$

$\therefore$ (4) has real and equal roots

$\therefore x = \frac{-b \pm \sqrt{D}}{2a}$

$= \frac{-(-40) \pm \sqrt{0}}{2(1)}, \frac{40}{2}, 20$

$x=20$

$y=40-20=20$

Thus, length and breadth are 20m

Hence, the given rectangular park exists and it is a square.

## Extra Questions (HOT)

Q1 $\frac{1}{x} + \frac{1}{x+1} = \frac{4}{1}$, find the value of x.

Solu $\Rightarrow \frac{x+1+x}{(x)(x+1)} = \frac{4}{1}$

$\Rightarrow \frac{2x+1}{x^2+x} = \frac{4}{1}$

$\Rightarrow 4x^2+4x = 2x+1$

$\Rightarrow 4x^2+2x = 1$

$\Rightarrow 4x+2x-1 = 0$

$a=4, b=2, c=-1$

$D = (2)^2 - 4(4)(-1)$

$= 4+16 = 20 > 0$

Roots are real

$x = \frac{-b \pm \sqrt{D}}{2a}$

$x = \frac{-2 \pm \sqrt{20}}{2\times 4}$

$x = \frac{-2 \pm \sqrt{20}}{8}$

$x = \frac{-2+2\sqrt{5}}{8}$ or $\frac{-2-2\sqrt{5}}{8}$

Q2 i) $\frac{1}{a} + \frac{1}{b} + \frac{1}{x} = \frac{1}{a+b+x}$

ii) $\frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} = 1$

iii) $\frac{x-2}{x-4} - \frac{x+4}{x-2} = 4$

ii) Solu $\Rightarrow \frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} = 1$

$\Rightarrow \frac{1}{(x-2)}\left[\frac{1}{(x-1)} + \frac{1}{(x-3)}\right] = 1$

$\Rightarrow \frac{1}{(x-2)} \cdot \left[\frac{x-3+x-1}{(x-1)(x-3)}\right] = 1$

$\Rightarrow \frac{1}{(x-2)}\left[\frac{2x-4}{(x-1)(x-3)}\right] = 1$

$\Rightarrow \frac{1}{(x-2)}\left[\frac{2(x-2)}{(x-1)(x-3)}\right] = 1$

$\Rightarrow \frac{1}{(x-1)(x-3)} = 1$

$\Rightarrow 1 = (x-1)(x-3)$

$\Rightarrow x^2-3x-x+4 = 0$

$\Rightarrow x^2-4x+1 = 0$

~~$x^2-3x-x+1=0$~~ ~~Ans~~

~~$x(x-3)-1(x-3)$~~

??

i)-sol ⇒ $\frac{1}{a}+\frac{1}{b}+\frac{1}{x}=\frac{1}{a+b+x}$

⇒ $\frac{1}{a}+\frac{1}{b}=\frac{1}{a+b+x}-\frac{1}{x}$

⇒ $\frac{(b+a)}{ab}=\frac{x-(a+b+x)}{(a+b+x)(x)}$

⇒ $\frac{b+a}{ab}=\frac{x-a-b-x}{(a+b+x)(x)}$

⇒ $\frac{(a+b)}{ab}=\frac{-(a+b)}{(a+b+x)(x)}$

⇒ $\frac{1}{ab}=\frac{-1}{ax+bx+x^2}$

⇒ $ax+bx+x^2=-ab$

⇒ $ax+bx+x^2-ab=0$

⇒ $x^2+(a+b)x+ab=0$

⇒ $(x+a)(x+b)=0$

⇒ $x=-a, x=-b$

iii)-sol ⇒ $\frac{x-2}{x-4}-\frac{x+4}{x-2}=4$

Let $\frac{x-2}{x-4}$ be $t$

⇒ $t-\frac{1}{t}=4$

⇒ $\frac{t^2-1}{t}=4$

⇒ $t^2-1=4t$

⇒ $t^2-4t-1=0$

$a=1, b=-4, c=-1$

$D=(-4)^2-4(1)(-1)$

$=16+4=20>0$

Roots are real

$t=\frac{-b\pm\sqrt{D}}{2a}=\frac{-(-4)\pm\sqrt{20}}{2(1)}$

$=\frac{4\pm2\sqrt{5}}{2}=\frac{2(2\pm\sqrt{5})}{2}$

$=2\pm\sqrt{5}$

⇒ $\frac{x-2}{x-4}=2\pm\sqrt{5}$

⇒ $x-2=(2\pm\sqrt{5})(x-4)$

⇒ $x-2=2x-8\pm4\sqrt{5}\pm x\sqrt{5}$

⇒ $2x-x-8+2\pm4\sqrt{5}\pm x\sqrt{5}=0$

⇒ $x-6\pm\sqrt{5}(x+4)=0$

⇒ $x\pm\sqrt{5}(x+4)=6$

⇒ $x=6\pm\sqrt{5}(x+4)$

28/3/25